Hirzebruch–Riemann–Roch theorem

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Contents

I will try to have the introduction have as few prerequisites as possible so that what I'm doing can at least be explainable, but it will be inevitable to bring up scheme theory and k-theory in the paper so I will just drop the pretext of accessibility at that point.

A natural question to ask in complex and algebraic geometry is "how many holomorphic/meromorphic/algebraic/analytic/etc functions are there on a (suitable) space X". By letting $\mathcal{F}(X)$ represent our choice of functions (the global sections of a [quasi-coherent] sheaf $\mathcal{F}$), we shall denote this question a question about finding information about the 0th sheaf cohomological group

$$H^0(X, \mathcal{F})$$

which by [1] is equal to $\mathcal{F}(X)$. This group shall sometimes be trivial, and the reason for its trivialness can be described by higher order sheaf cohomology groups:

$$H^1(X, \mathcal{F}), H^2(X, \mathcal{F}), \ldots$$

exactly how to extract this information is known as the *Riemann Roch Problem*. To motivate the direction the problem took, let us answer this question in some simple cases.

The simplest case would be to deal with polynomial functions over $X = \mathbb{C}$ which are determined by a finite set of roots (counting multiplicity) and a constant:

$$p(z) = c(z - a_1) \cdots (z - a_n)$$

and similarly for rational functions. For future connection to sheaves let $\mathcal{O}_X(X)$ represent all algebraic functions on X and $\mathcal{M}_X(X)$ represent all rational functions on X; other common notation is $\mathbb{C}[X] = \mathcal{O}_X(X)$ and $\mathbb{C}(X) = \mathcal{M}_X(X)$ where $\mathbb{C}$ is often replaced with some other field K or ring R. If we ask for holomorphic functions so that $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{H}(\mathbb{C}))$, then, then the Weierstrass Factorization Theorem (see [8, section 4.2]) tells us that $f \in \mathcal{H}(\mathbb{C})$ factors as an infinite polynomial along with some correction factors

$$z^m e^{g(z)} \prod_k^\infty \left[\left(1 - \frac{z}{a_k}\right) e^{p_k(z)} \right]$$

where

$$p_k(z) = \frac{z}{a_k} + \frac{1}{2}\left(\frac{z}{a_k}\right)^2 + \cdots + \frac{1}{m_k}\left(\frac{z}{a_{m_k}}\right)^m$$

is the correction factor and the a_k are accumulation points; in particular there are analytic concerns. A meromorphic function f on $\mathbb{C}$ can be determined by two entire function $f = g/h$ (again see [8, section 4.2]); this is possible as for poles of f we can define a function h with the appropriate roots and multiplicities. In other words, determining meromorphic functions becomes a classification of non-accumulating sequences in $\mathbb{C}$. The situation requires even more analytic information in hyperbolic space D^2 (think of the existence of $\sum_n z^{n!}$ and $1/(1-z)$). To understand the algebraic behaviour, limiting the scope to *compact spaces* such as Riemann surfaces or projective varieties gets much better as any holomorphic/meromorphic functions must have finitely many roots/poles by the Identity theorem (that if two functions agree on a set of accumulation points, they are equal), and so we can focus our efforts within the algebraic setting. This setting is plenty rich enough for study and often offers a good starting point to generalize to the non-compact case[1]. In fact, by the efforts of Serre, Chow, and Grothendeick, the category of analytic functions on (suitable) compact spaces is equivalent to the category of algebraic functions on these spaces ([9], [4]), meaning compact spaces are "intrinsically" algebraic. We shall prove some of the pertinent parts of this correspondence in ref:HERE.

Let us try finding holomorphic/meromorphic functions on $X = \mathbb{C}\mathrm{P}^1$ and see if they are determined by the points of $\mathbb{C}\mathrm{P}^1$. Immediately, $\mathbb{C}\mathrm{P}^1$ has only constant holomorphic functions by an easy generalization of Liouville or the maximum modulus principle on manifolds. By the Residue Theorem, the sum of the number of roots and poles of a meromorphic function must be 0.

Let us represent this formally. Let

$$D = \sum_i n_i p_i \qquad p_i \in \mathbb{C}\mathrm{P}^1$$

be a finite formal sum; we may think of it as the free (abelian) group $F^{\mathbb{Z}}(\mathbb{C}\mathrm{P}^1)$. Call D a *divisor*, and let let $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$ represent this group. A divisor can be thought of as an object that remembers which roots/poles with which multiplicity we are interested in. Define:

$$\mathrm{div}(f) = \sum_i \nu_{p_i}(f) p_i$$

where $\nu_{p_i}(f)$ is the order of the zero/pole, and is negative if it is a pole. This forms a subgroup of $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$; represent it as $\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$ ("P" for principal), and elements of this subgroup are called *principal divisors*. This subgroup represents the divisors that are *representable* as a meromorphic function. Let us find which divisors can be represented. Let:

$$\deg D = \sum_i n_i$$

Notice that $\deg(-)$ is certainly a group homomorphism. By our earlier remark:

$$\deg(\mathrm{div}(f)) = 0$$

which is a restriction applied to any meromorphic function on $\mathbb{C}\mathrm{P}^1$. It is easy to see that this restriciotn would work on any compact Riemann surface (see [8, chapter 6]). If $\deg(D) = 0$, by

[1] Though these cases are still very much being studied, see ref:HERE

our earlier remark we have that there exists an f such that $\text{div}(f) = D$, showing this conditions characterizes meromorphic functions $f \in \mathcal{M}(\mathbb{C}\text{P}^1)$ (but it shall not on other Riemann surfaces). Now, consider the *(Weil) divisor class group*

$$\text{Cl}(\mathbb{C}\text{P}^1) = \frac{\text{Div}(\mathbb{C}\text{P}^1)}{\text{PDiv}(\mathbb{C}\text{P}^1)}$$

for any $[D] \in \text{Cl}(\mathbb{C}\text{P}^1)$, $\deg([D])$ is well-defined as $\deg(D) = \deg(D + f)$ for any $f \in \text{PDiv}(\mathbb{C}\text{P}^1)$. If $\deg(D) = \deg(D')$, then $\deg(D) - \deg(D) = \deg(D - D') = 0$, and hence $D - D' = \text{div}(f)$. Therefore, we have that:

$$\text{Cl}(\mathbb{C}\text{P}^1) \cong \mathbb{Z}$$

which can be thought of as measuring the obstruction in a meromorphic function being defined by its points/roots. As sheaf cohomology measures obstruction of local information having global information, we shall have:

$$\text{Cl}(\mathbb{C}\text{P}^1) \cong H^1(\mathbb{C}\text{P}^1, \mathcal{O}^*_{\mathbb{C}\text{P}^1})$$

the details of this isomorphism and why $\mathcal{O}^*_{\mathbb{C}\text{P}^1}$ shall be given in ref:HERE.

Translating the result for $(\mathbb{C}, \mathcal{M}_{\mathbb{C}})$, we see that:

$$\text{Cl}(\mathbb{C}) = \frac{\text{Div}(\mathbb{C})}{\text{PDiv}(\mathbb{C})} \cong 0$$

as every divisor can be represented by a principal divisor. We can interpret this as there being no impediments to using local information to construct meromorphic functions, and we shall later also see that all cohomology groups shall be trivial. Notice that the ring $\mathbb{C}[x]$ is a UFD; this algebraic fact shall be given geometric meaning in ref:HERE.

Let us look at the next simplest nontrivial example. Let $X = \mathbb{C}/\Lambda$ where $\Lambda \subseteq \mathbb{C}$ is a lattice spanned by the $\mathbb{R}$-linearly independent elements $(1, \tau)$ and let $\mathcal{F} = \mathcal{M}_X$ (all rational functions on X); X is called an *elliptic curve*, and so we shall label it as E. As mentioned earlier, if $f \in \mathcal{M}_E$, $\text{div}(f) = 0$. However, if $\deg(D) = 0$, this does not imply there exists an $f \in \mathcal{M}_E$ such that $\text{div}(f) = D$. To see this, let analyze $\mathcal{M}_E$ a little more closely. As $\mathbb{C}/\Lambda$ is the quotient by Λ, any meromorphic function must respect the Λ structure:

$$f(z + n + m\tau) = f(z) \qquad \forall n + m\tau \in \Lambda$$

such a function is called *doubly periodic* or *elliptic function* (as they are defined on elliptic curves). Using this periodic condition, it is simple to see that $\text{div}(f) = 0$ by looking at:

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f'(z)}{f(z)} dz = 0$$

where D is the fundamental domain[2], namely the opposing edges cancel out. We can modify this integral to find other properties, for example:

$$\frac{1}{2\pi i} \int_{\partial D} z \frac{f'(z)}{f(z)} dz = n + m\tau \in \Lambda$$

[2] If there are zero/poles on the path, we simply shift around appropriately

which is nonzero as z is not doubly periodic. Replacing z by other algebraic functions all end up in Λ, and so we see that if $D = \operatorname{div}(f)$:

$$\sum_i n_i p_i \equiv 0 \text{ mod } \Lambda$$

where here $\sum_i n_i p_i \notin F^{\mathbb{Z}}(E)$ but $\sum_i n_i p_i \in E$, namely we are trying p_i as a complex number multiplied by n_i and summed. Certainly there is no such restriction on a general divisor $D \in \operatorname{Div}(E)$ of degree 0. We thus have shown that there is a surjective map:

$$\operatorname{Cl}^0(E) = \frac{\operatorname{Div}^0(E)}{\operatorname{PDiv}(E)} \twoheadrightarrow \mathbb{C}/\Lambda \tag{1}$$

where $\operatorname{Div}^0(E)$ are all divisor of degree 0. It is in fact an isomorphism, which comes from constructing an elliptic functions using the Weierstrass $\wp$-function, the details are in [8, chapter 5]. Thus, we get the short exact sequence:

$$0 \to \mathbb{C}/\Lambda \to \operatorname{Div}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

often called the *Jacobi-Abel* exact sequence as $\mathbb{C}/\Lambda$ is a Jacobian variety, and a short exact sequence:

$$0 \to \operatorname{Cl}^0(E) \to \operatorname{Cl}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

that together fit into the commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Div}^0(E) \arrow[r, hook] \arrow[d, two heads, "\pi"] & \operatorname{Div}(E) \arrow[r, "\deg"] \arrow[d, two heads, "\pi"] & \mathbb{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \operatorname{Cl}^0(E) \arrow[r, hook] & \operatorname{Cl}(E) \arrow[r, "\deg"] & \mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Continuing on to other Riemann surfaces, we shall loose the isomorphism given in equation (1) and is the beginning of the study of Jacobian varieties (see [8, section 6.4]), while going in higher dimensions is the beginning of the study of Weil and Cartier Divisors (see [1]), and study how far away a space is from being constructed by "principal" objects. Overall, the question becomes less about finding which functions exist and more about some global properties of the space X.

To answer the Riemann Roch problem, a more granular approach to find which functions (of various kinds) exist on a space X must be taken. As divisors keep track of root/pole information, they are a great tool in classifying functions that "satisfy" the properties of the divisor. To make this precise, define an order on divisors $\operatorname{div}(X)$ where (for now) X is either a Riemann surface or a smooth projective curve. Define a partial order on $\operatorname{Div}(X)$ by saying $D_1 \geq D_2$ if each coefficient of $D_2 - D_1$ is non-negative. Then we can consider:

$$\mathcal{O}_X(D)(X) = \{f \in \mathcal{M}_X(X)^* \mid \operatorname{div}(f) + D \geq 0\}$$

In other words, if $D = \sum_i n_i p_i$, then $\mathcal{O}_X(D)(X)$ is all meromorphic functions f where it has a pole at p_i of order at most n_i if $n_i > 0$ and roots at p_i of order at least $|n_j|$ if $n_j < 0$ (and $\mathcal{O}_X(D)$ is a [coherent] sheaf). From this, we can reformulate the Riemann Roch problem to a question about $H^0(X, \mathcal{O}_X(D))$. The famous Riemann-Roch theorem as proven by Riemann and Roch (where Riemann proved the Riemann-inequality, and Roch provided the necessary work to make it an equality, see [6]) is now known as *Riemann Roch for curves* and is the equality:

$$\dim H^0(X, \mathcal{O}_X(D)) - \dim H^1(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g \tag{2}$$

where g is the genus of the Riemann surface or smooth projective curve (see [10] or [1] for defining the genus of a curve via scheme theory, or for a direct computational proof using normalization see [8]). In the original phrasing of the Riemann Roch theorem, the group $H^1(X, \mathcal{O}_X(D))$ is replaced with $H^0(X, \mathcal{O}(K_X - D))^*$ where K_X is the canonical divisor of X; these groups are isomorphic by Serre Duality (see [1, section 6.3]). Then this space is identified with the space of meromorphic 1-forms (either through some simple techniques from complex analysis, see Brill-Noether Reciprocity in either [6] or [8], or by combining of Poincaré duality with DeRham's Theorem, see either [3] or [7]). Switching to the traditional notation $l(D) = \dim H^0(X, \mathcal{O}_X(D))$ and $i(D) = \dim H^0(X, \mathcal{O}_X(K_X - D))^*$ we get:

$$l(D) - l(K - D) = \deg(D) + 1 - g$$

idk if I will prove RR for curves or relegate it to a book. As we phrased the Riemann-Roch problem in the beginning, this formula would then be asking for the value of $l(D)$. There are cases where $l(K - D)$ disappears (for example if $D = 0$ or $D = K$), and there are many cases we can compute. It is simple to compute the right hand side but in general it is harder to computer the left hand side. Thus, this formula becomes a new constriction that helps us find $l(D)$. There are a set of theorems known as *vanishing theorems* that specify under which conditions $l(K-D)$ (or more generally, higher cohomology groups) vanish (see [5]); under these conditions we exact recover $l(D)$.

Let us work towards the intuition behind the generalization. The notation on left hand side in equation (2) can be compactified by using the Euler-Poincaré characteristic (or simply the Euler characteristic): $\chi(X, \mathcal{O}_X(D)) = \sum_{i=0}^{\infty} (-1)^i \dim H^i(X, \mathcal{O}_X(D))$ so that:

$$\chi(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g$$

Note that $H^i(X, \mathcal{O}_X(D)) = 0$ for $i > 1$, cite? Reference for later? giving the equality. The right hand side shall be drastically different, split into special invariants of vector bundles known as the Chern class and Todd class mention the "duality" between them? see notes . In fact, the Euler Characteristic shall also turn out to be an invariant on vector bundles, and hence the Riemann Roch theorem generalizes to a question about the existence of sections on vector bundles over a compact manifolds. We shall mention the generalization to smooth quasi-projective schemes over a field near the end, and also talk a bit about a recent paper ([2]) generalizing to finite dimensional Noetherian schemes.

To motivate how vector bundles will enter the picture, it is worth-while seeing how $\mathcal{O}_X(D)$ can be interpreted as a line bundle. To that end, recall that the projective curve $X = \mathbb{P}^1$ as defined in classical algebraic geometry only has constant global sections. Namely, if $U_0, U_1 \cong \mathbb{A}^1$ is the usual cover with global sections (i.e. functions) $\mathcal{O}_X(U_0) \cong \mathcal{O}_X(U_1) = K[x]$ and the compatibility condition on $U_0 \cap U_1$ is $f(z) = g(1/z)$, namely:

$$\begin{tikzcd} U_0 & U_1 & & \mathbb{A}^1 & \mathbb{A}^1 \\ U_0 \cap U_1 \arrow[u, hook] \arrow[r, dash, "\cong"] & U_0 \cap U_1 \arrow[u, hook] & \text{i.e.} & \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \arrow[r, dash, "\cong"] & \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \end{tikzcd}$$

gives:

$$\begin{array}{ccccc} \mathcal{O}_X(U_0) & & \mathcal{O}_X(U_1) & & K[x] & & K[y] \\ \downarrow & & \downarrow & \text{i.e.} & \downarrow & & \downarrow \\ \mathcal{O}_X(U_0 \cap U_1) & \xrightarrow{\cong} & \mathcal{O}_X(U_0 \cap U_1) & & K[x, 1/x] & \xrightarrow{\cong} & K[y, 1/y] \end{array}$$

where $x \mapsto 1/y$ and $1/x \mapsto y$, Then if $F([x' : y']) \in \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$ is a global function, then $F([x : 1]) \to f(x) \in \mathcal{O}_X(U_0)$ and $F([1 : y]) \to g(y) \in \mathcal{O}_X(U_1)$ are the restrictions of the function to U_0 and U_1 respectively, and by the above diagram it must be that these functions map to the same element in the restriction, namely:

$$f(z) \to f(x) = f(1/y) = g(y) \leftarrow g(y)$$

but it is impossible that $f(1/y) = g(y)$ unless f, g are the same constant. Thus, it must be that the only global functions on the projective line are constants:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = K$$

The fundamental limitation here is that the restriction $f(x) = g(1/x)$ is very strong. What if we can loosen this restriction? For example what if

$$f(x) = xg\left(\frac{1}{x}\right)$$

Then we can in fact form linear homogeneous functions! For example consider

$$F([x' : y']) = ax' + by'$$

Then F maps to $f(x) = ax + b$ and $g(y) = a + by$. Then:

$$f(x) = x\left(a + b\frac{1}{x}\right) = ax + b = xg\left(\frac{1}{x}\right)$$

which works! More generally, if the condition $f(x) = x^n g(1/x)$ allows for degree n homogeneous polynomials. Denote the degree n-homogeneous polynomials by $\mathcal{O}_{\mathbb{P}^1}(n)(\mathbb{P}^1)$.

also mention how the weil/cartier divisor stuff earlier isomorphic to line bundles, hence that discussion really about properties of line bundles.

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